Rakib Mahmud

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Education

- **Khulna University of Engineering & Technology (KUET)** *November 2018 - March 2024*
  B.Sc. in Computer Science and Engineering Khulna, Bangladesh
  - CGPA: 3.71/4.00

Research Interests

- Artificial Intelligence
- Computer Vision
- Federated Learning
- Deep Learning
- Natural Language Processing
- Machine Learning and Unlearning

Research Experience

- **Federated Unlearning Is Not Fail-Safe: A Measurement Study of Verifying, Attributing and Recovering Forgetting Under Client Faults** *2026*
  Advisor: *Dr. A. B. M. Alim Al Islam, Dept. of Computer Science & Engineering, BUET*
  - Measured how crash, value, and Byzantine client faults break federated unlearning across four datasets; robust aggregation left up to 19.8 points more backdoor than FedAvg.
  - Showed that trigger-based checks can certify deletions never carried out; built a verify–attribute–recover loop that restores a destroyed model from 13.0% to 76.1% accuracy.
- **Multi-Class Chest Disease Classification from X-Rays Using PCA-Enhanced Vision Transformer** *2024 - 2025*
  Advisor: *Dr. Ch. Md. Rakin Haider, Dept. of Computer Science & Engineering, BUET*
  - Developed a deep learning framework for multi-class chest disease classification (COVID-19, Pneumonia, Tuberculosis, Normal) from X-ray images, using PCA to reduce feature redundancy and computational cost while preserving discriminative information.
  - Implemented a hybrid ensemble of Vision Transformer (locality-aware self-attention) and EfficientNet to capture global and local radiographic patterns, with Grad-CAM and LIME for clinical interpretability.
- **A Hybrid Deep Learning Approach with Explainable AI for Bangladeshi Birds Classification** *2023 - 2024*
  Thesis Supervisor: *Dr. Sk. Md. Masudul Ahsan, Dept. of Computer Science & Engineering, KUET*
  - Built a pipeline using YOLOv8 for bird localization and transfer learning for classification on a self-curated dataset.
  - Designed a hybrid CNN with Lab color space histogram features for limited computational resources, and integrated Grad-CAM to enhance interpretability for biodiversity applications.

Teaching Experience

- **Bangladesh University of Business and Technology (BUBT) [🌐]** *July 2025 - Present*
  Lecturer, Dept. of Computer Science & Engineering Dhaka, Bangladesh
  - Taught undergraduate courses and labs on Data Structures, Software Engineering, and Pattern Recognition.
  - Advised and mentored undergraduate cohorts; served as Course Coordinator for Pattern Recognition.
- **Dhaka International University (DIU) [🌐]** *July 2024 - July 2025*
  Lecturer, Dept. of Computer Science & Engineering Dhaka, Bangladesh
  - Taught courses and labs on Discrete Mathematics, Structured and Object-Oriented Programming, Compiler Design, Computer Architecture, Markup and Scripting Languages, and Software Engineering.
  - Served as Batch Counselor (meeting documentation, academic tour coordination, research publication assistance) and Co-Moderator of the DIU Film and Photography Club.

Publications

[1] Ankan Saha, **Rakib Mahmud**, and Rifat Shahriyar. **"Deconstructing the Garbage-First Collector: A Python Simulation Study with an Adaptive Hot-Card Threshold."** *4th International Conference on Computing Advancements (ICCA 2026)*. ACM, 2026. *(Accepted)*

[2] **Rakib Mahmud** and Sk Md Masudul Ahsan. **"A Hybrid Deep Learning Approach with Explainable AI for Bangladeshi Birds Classification."** *2025 2nd International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*. IEEE, 2025.

[3] Asif Mohammed Saad, **Rakib Mahmud**, and Sunanda Das. **"Forecasting Crucial Biogeochemical Indicators of the Southern Ocean for Climate Monitoring Using Modified Kernel-Based Support Vector Regression."** *International Conference on Big Data, IoT and Machine Learning*. Singapore: Springer Nature Singapore, 2023.

PROJECTS

- **End-to-End Sentiment Analysis System Using Traditional ML and Transformer Models**
  Python, PyTorch, OpenCV, Scikit-Learn, NumPy, Pandas, Matplotlib [GitHub]
  * Built a sentiment analysis pipeline on a large-scale Reddit dataset, combining TF-IDF bigram features and class-imbalance handling with BERT classification for improved recall, with MLflow experiment tracking.
- **KUETPS Website Using PHP**
  PHP, MS SQL, Bootstrap, CSS, HTML [GitHub]
  * Created KUETPS Photographic Society website with user authentication, image interaction (upload, like, dislike, comment), secure database storage, and profile customization.
- **Automobile Company's Database**
  SQL, Oracle [GitHub]
  * Designed and implemented a comprehensive database system for automobile company operations with normalized schema, complex queries, and transaction management.
- **Custom Programming Language**
  Flex, Bison [GitHub]
  * Developed a sophisticated compiler integrating lexical and syntax analysis, efficient code optimization and generation, robust error handling, and cross-platform compatibility.
- **Dots and Boxes Game**
  Python [GitHub]
  * Developed a dynamic Dots and Boxes game for human-vs-AI gameplay with Alpha-Beta Pruning for AI decision-making optimization, enhancing player experience.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Java
- **Data Analysis & Visualization:** OpenCV, Scikit-Learn, NumPy, Pandas, Matplotlib
- **Databases:** Oracle, Microsoft SQL Server, Firebase
- **Web Programming:** HTML, CSS, JavaScript, PHP
- **Software Tools:** MS Word, Excel, PowerPoint
- **Developer Tools:** Git, GitHub, OpenGL, LaTeX

HONOURS AND AWARDS

- **Dean's Award** *2018-19 and 2019-20*
  Khulna University of Engineering & Technology [Web]
  * Awarded for Freshman and Sophomore Years
- **Dutch-Bangla Bank Scholarship** *2019-2024*
  Dutch-Bangla Bank Foundation [Web]
  * Awarded for Eight Consecutive Semesters

EXTRACURRICULAR ACTIVITIES

- **President** *2023 - 2024*
  KUET Photographic Society (KUETPS) [Web]
  * Led the creative photography community of KUET that nurtures visual storytelling through workshops, photo walks, training sessions, exhibitions, and competitions for photographers of all levels.
- **Member Secretary & Founding Member** *2023 - 2024*
  CSE KUET Research Society (KRS) [Web]
  * Contributed to KUET's first research-focused club by organizing workshops and seminars, mentoring students on research and academic writing.

REFERENCES

Dr. Sk. Md. Masudul Ahsan
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Department of Computer Science & Engineering
Khulna University of Engineering & Technology (KUET)
Email: smahsan@cse.kuet.ac.bd

Sunanda Das
Assistant Professor
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